PATTERNS IN LIFE: DIVERSITY AND CLASSIFICATION

Biodiversity: Introduction

- When we study the variety of **organisms** (living things) and **group** them by their features, many daily life observations become easy to understand.
- **Example 1:** Some plants can survive long **droughts** (long periods without rain), but others cannot. This depends on the structural features of the plant. For example, pine has thin needle-like leaves that lose less water, and its seeds have stored food.
- **Example 2:** Certain insects protect crops while others damage them. Insects like ladybird beetles (small, round, red beetles with black spots) eat the pests, and bees pollinate (carry pollen from one flower to another so that seeds can form) flowers, so they help the crop. Locusts (grasshopper-like insects that move in huge swarms and eat crops) and caterpillars (young ones of butterflies and moths) feed on the crop, so they damage it. We can tell them apart only if we identify and group them correctly.



Fig. 1: Organisms in four different habitats: icy polar region (polar bear, seals, whale), hot desert (camels, cactus, snake, lizard), dry grassland (deer, lion, eagle, monkey) and rainforest (elephant, hornbill, monkeys)

✓ **Biodiversity:**

- The Earth is home to an enormous variety of **life forms**.
- Some are **microscopic organisms** (so small that we cannot see them with the naked eye). Others are giant trees. There are glowing jellyfish and soaring (flying high) eagles.
- Life exists in countless forms and **habitats** (places where organisms live): from the snow-clad (covered with snow) Himalayas to the coral reefs (underwater rock-like structures made by tiny sea animals called corals) of the Andaman Sea.
- This immense (very large) variety of living organisms is called **biodiversity**.
- Biodiversity is essential for life on the Earth. Every organism plays a role in keeping nature stable and functioning.

✓ **Role of organisms in nature (interconnections):**

- **Microscopic algae** (simple green plants, mostly living in water) in the oceans release most of the oxygen we breathe.

- **Fungi and bacteria** (very small single-celled organisms without a true nucleus) **decompose** (break down) fallen leaves and convert waste into manure (natural fertiliser made from rotted plant and animal waste). This makes the soil fertile (rich in nutrients and good for growing plants).
- **Birds, bees and bats** pollinate flowers.
- **Plants** capture sunlight to prepare food. This food supports nearly all life on the Earth.
- These **interconnections** help sustain (keep going) **ecosystems** (a community of organisms living together with their non-living surroundings like air, water and soil) and make the Earth suitable for living organisms.
- All organisms depend on each other, because every group does a job that others need. So if one link is removed, the others also suffer.

✓ **How humans depend on biodiversity:**

- Humans depend on biodiversity for **food, shelter, medicines** and **livelihoods** (means of earning).
- For centuries, farmers used their practical knowledge to conserve diverse (of many different kinds) crop varieties with useful characteristics, such as **drought tolerance** (ability to survive with very little water), **pest resistance** (ability to fight insects and diseases that harm crops) and the ability to grow in **nutrient poor soils** (soils with very little food material for plants).
- Farmers realised that diversity reduces the risk of **crop failure** and strengthens **food security** (enough food available for everyone at all times).
- Diversity reduces the risk of crop failure because, with many varieties, at least some plants survive a drought or a pest attack. If only one variety is grown, one drought or one pest attack can destroy the whole crop.

✓ **Need to classify:**

- In such a vast diversity, organisms share certain **similarities** but also show many **differences**.
- To study diversity systematically, scientists **group and classify** organisms on the basis of their **shared characteristics** and **evolutionary relationships** (how they are related through evolution).
- Classification helps us understand how organisms are related and how they function.
- This knowledge is used in ecosystem management, biodiversity conservation (protection and saving), sustainable (that can go on for a long time without harming nature) farming, and so on.

India as a Biodiversity Hotspot

- **Diverse natural landscape of India:**
  - Mountains in the north
  - Desert in the west
  - Rainforests in North East India
  - Plateaus (high and flat areas of land) in the south
  - Long coastlines along the Arabian Sea and the Bay of Bengal
  - Each region has distinct **soil types** and different **climatic conditions** like temperature and rainfall.
  - These diverse habitats together support a wide variety of **species**.

✓ **Endemic species:**

- Some species are restricted to particular regions of the world and are **not found naturally anywhere else**. Such species are called **endemic species**.
- Examples found only in India: **Nilgiri tahr**, **Lion-tailed macaque** (a type of monkey), Indian variety of the pitcher plant ***Nepenthes khasiana*** and **Neelakurinji**.

Conceptual Learning

⇒ **Know these species:**

- **Nilgiri tahr:** A wild mountain goat found only on the high grassy hills of the Western Ghats in Tamil Nadu and Kerala. It is the state animal of Tamil Nadu.
- **Lion-tailed macaque:** A black monkey with a silver-white mane (long hair on the neck) around its face and a tuft (a small bunch of hair) at the end of its tail, like a lion. It lives only in the rainforests of the Western Ghats and is endangered (in danger of dying out).
- ***Nepenthes khasiana*:** A pitcher plant found only in the Khasi hills of Meghalaya. Its leaf forms a jug-like trap that catches and digests insects to get nitrogen.
- **Neelakurinji:** A shrub of the Western Ghats hills (like Munnar and the Nilgiris) with blue-purple flowers. It flowers only once in 12 years and turns whole hills blue.

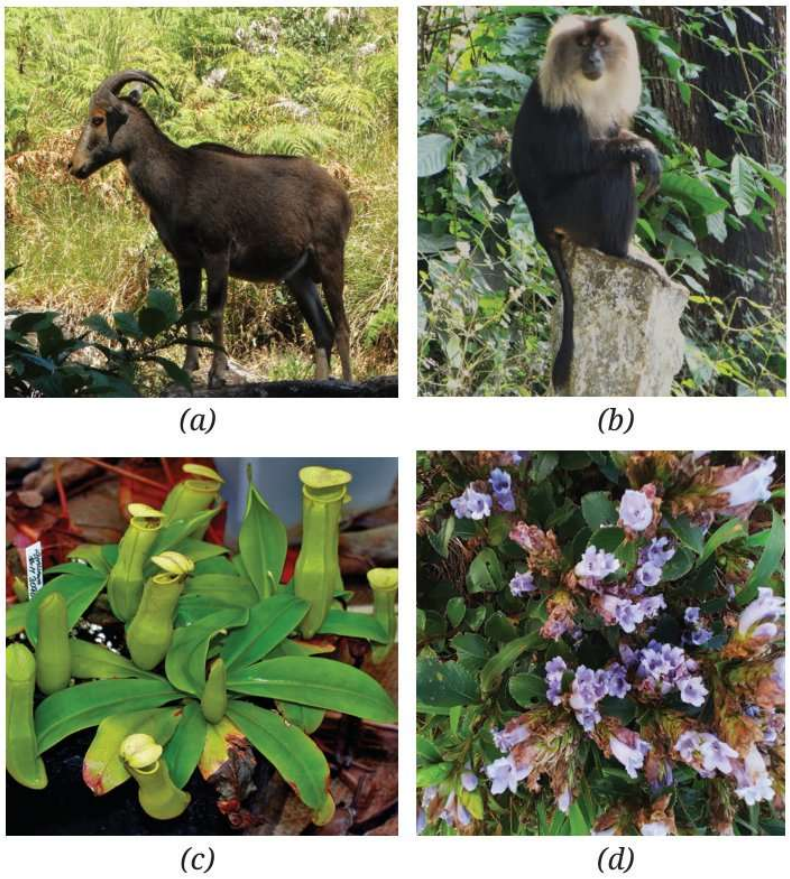


Fig. 2: Endemic species of India: (a) Nilgiri tahr, (b) Lion-tailed macaque, (c) Nepenthes khasiana (pitcher plant), (d) Neelakurinji

✓ **Biodiversity hotspots:**

- Regions that support a **large number of endemic species** and have undergone **significant habitat loss** (destruction of the natural home of organisms) are called **biodiversity hotspots**.
- These areas are particularly important for **biodiversity conservation**.
- Examples of global biodiversity hotspots: **Western Ghats**, **Indo-Burma** (includes North East India), **the Himalayas** and **Sundaland** (includes the Nicobar Islands).
- These areas are especially rich in number and in the diversity of organisms.
- Protecting them is important, as they support **food webs** (many food chains linked with each other) and help ecosystems remain healthy.
- Hotspots need special protection because they have many endemic species (found nowhere else) and these species are already in danger due to heavy habitat loss. If such a place is destroyed, those species are lost from the whole world.

How has Biodiversity Evolved?

➢ The biodiversity that we see today on the Earth was not always the same.

➢ **Small differences among individuals** affected their chances of **survival and reproduction** by helping them **adapt** (adjust) to changed conditions.

➢ These differences **accumulated** (added up) over many generations (one stage of parents and their offspring) and gave rise to **new forms of life**.

➢ So, today's diversity is an outcome of continuous changes over a vast span of time, shaped by interactions between organisms and their surroundings.

➢ Classification provides a **systematic** (orderly) **framework** (an orderly method) to study this biological diversity.

⇒ **Sangam Tinai and sacred groves: ancient Indian understanding of habitats:**

- Old Indian traditions like the ***Sangam Tinai*** (grouping of landscapes in old Tamil literature) and the protection of **sacred groves** (small patches of forest protected by local people for religious reasons) show a deep literary and cultural understanding of landscapes and their **biota** (living organisms of an area).
- These customs preserved locally diverse habitats. This matches present-day **ecological principles**, even though it was never written as a formal ecological theory.

How to Classify Organisms?

➢ The same organism can fit into different groups depending on the **criterion** (basis) we choose. So scientists need a proper, **systematic** way of organising the Earth's living diversity. This is called **classification**.

➢ Scientists do not group organisms only by habitat or habit, because then unrelated animals come together (the bat and the eagle both fly).

✓ **Some criteria to classify living organisms:**

- Scientists first look at broad and easily visible features, and then at more detailed features. Some of the criteria (bases for grouping; plural of criterion) are:

1) **External features:** visible characteristics, such as shape, size and body organisation (arrangement of cells, tissues and organs in the body).
2) **Mode of nutrition: autotrophic** (make their own food) or **heterotrophic** (depend on others for food).
3) **Internal structures:** skeletal patterns, presence or absence of organs, and different types of tissues (groups of similar cells doing the same job).
4) **Cell structure: unicellular** (one cell) or **multicellular** (many cells); **eukaryote** (has a true, membrane-bound nucleus) or **prokaryote** (no true nucleus); presence or absence of cell wall.
5) **Ecological role:** producer, consumer or decomposer.
6) **Reproduction:** asexual and/or sexual methods.
7) **Genetic** (related to genes and DNA) **similarity:** similarities in inherited (passed on from parents to offspring) features, which scientists study in detail using **DNA** (the genetic material in cells that carries instructions from parents to offspring).

- These features help in understanding the similarities and differences among organisms.
- **Similar features in organisms suggest that they have evolved** (developed slowly over a long time) **from common ancestors** (earlier forms of organisms from which present ones have come)**.**

Conceptual Learning

The Need for Classification

➢ **Library example:** Suppose thousands of books are lying all over the floor of a big library. Finding one book, say *On the Origin of Species* by Charles Darwin, would be very hard. If the books are arranged by subject, author or section, we can find it easily.

➢ In the same way, the Earth is home to **millions of organisms**. Classifying them systematically helps us understand them better.

✓ **Biological classification:**

- Many organisms show similarities in their external features, internal structure (tissues and organs) and **cellular organisation**.
- Scientists group organisms on the basis of their similarities and/or differences. This organises information, shows relationships among organisms and helps us study life in a systematic manner.
- This scientific way of grouping living organisms is called **biological classification**.
- **Biological classification helps us in various ways:**
  1) It makes the study of living organisms more organised and systematic.
  2) It helps us understand the similarities and differences among living beings.
  3) It helps us understand how different organisms are related to one another and how they interact.
  4) It helps in identifying and naming newly discovered organisms.
  5) It supports biodiversity conservation by identifying organisms that are under the threat of **extinction** (dying out completely).
  6) It allows scientists all over the world to discuss organisms using a common system.

Biological Classification Systems Over Time

➢ Scientists have developed different classification systems over time, based on the knowledge available.

- **(a) Aristotle: artificial system (4th century BCE):**
  - He grouped animals on the basis of their **habitat**: land, water and air. He also grouped them by their **external appearance**.
  - **Limitation:** it relied mainly on easily observable external characteristics. Example: a whale and a fish both live in water, but they are very different animals.

  ⇒ **Classification of animals in the Rigveda and the Brihat Samhita:**
  - The *Rigveda* and the *Brihat Samhita* classify animals on the basis of their **habitat** (terrestrial: land, aquatic: water and aerial: air), **behaviour patterns** and **ecological roles**.
- **(b) Two kingdom system: Carolus Linnaeus (1758, 18th century):**
  - All living organisms were divided into Plantae and Animalia.
  - **Plantae:** organisms that do not move from one place to another and **synthesise** (make) their own food.
  - **Animalia:** organisms that move from one place to another and depend on other organisms for food.
  - **Problem:** this system created confusion about where to place *Amoeba*, *Paramecium* and bacteria. *Amoeba* and *Paramecium* move like animals, but they are **unicellular** and **heterotrophic**, while plants and animals are **multicellular** organisms.
- **(c) Three kingdom system: Ernst Haeckel (1866):**
  - A third kingdom (the largest group in classification), **Protista**, was added to include **unicellular microscopic organisms**.

Conceptual Learning

- **(d) Four kingdom system: Herbert F. Copeland (1938):**
  - As microscopes improved, scientists noticed that an *Amoeba* has a **true nucleus (membrane-bound)**, but bacteria do not.
  - Both are unicellular but very different. So bacteria were placed in a separate kingdom called **Monera**.
  - The four kingdoms were Plantae, Animalia, Protista and Monera. Protista now included mostly unicellular organisms **with a true nucleus**.
- **(e) Five kingdom system: Robert H. Whittaker (1969):**
  - Fungi like mushrooms do not move, like plants, but they have a **heterotrophic mode of nutrition**.
  - They obtain nutrients by **absorption** (taking in). Many absorb nutrients from **dead and decaying** (rotting) **matter**. Some are **symbiotic** (live in close association with other organisms) and some are **parasitic** (living on or inside another organism and taking food from it).
  - So fungi were placed in a separate kingdom. This gave five kingdoms: **Monera, Protista, Fungi, Plantae and Animalia**.

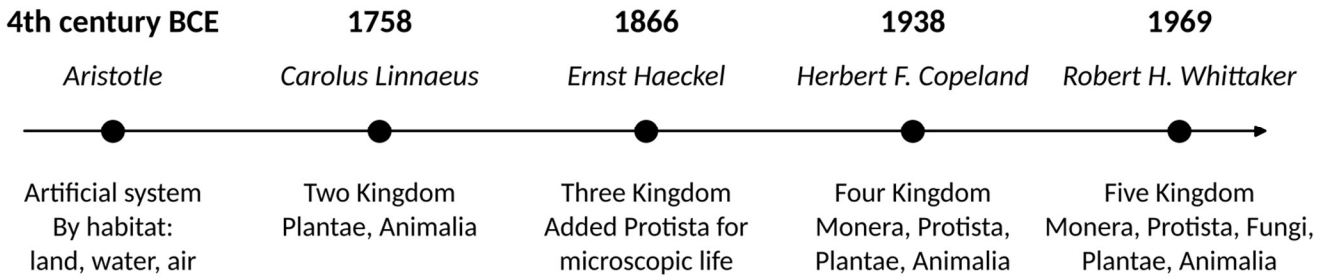


Fig. 3: Timeline of classification systems

- Classification systems kept changing because each new finding (unicellular organisms, true nucleus, mode of nutrition of fungi) did not fit the old groups. So classification changes as our knowledge and tools improve.

Five Kingdom Classification

➢ In the five kingdom classification, all life forms are grouped on the basis of certain common features.

➢ The five kingdoms are made on the basis of four criteria: **cell type** (prokaryote or eukaryote), **level of organisation** (unicellular or multicellular), **cell structure** (presence or absence of cell wall) and **mode of nutrition** (autotrophic or heterotrophic).

Table C: The five kingdoms at a glance

Feature	Monera	Protista	Fungi	Plantae	Animalia
Cell type	Prokaryotic	Eukaryotic	Eukaryotic	Eukaryotic	Eukaryotic
Level of organisation	Unicellular	Unicellular	Mostly multicellular	Multicellular	Multicellular
Cell wall	Present (rigid cell covering)	May or may not be present; if present, made of cellulose (a tough substance found in the cell walls of plants)	Present, made of chitin (a hard and tough substance, also found in the outer covering of insects)	Present, made of cellulose	Absent
Mode of nutrition	Autotrophic or heterotrophic	Autotrophic or heterotrophic	Heterotrophic (absorption)	Autotrophic	Heterotrophic

Feature	Monera	Protista	Fungi	Plantae	Animalia
Ecological role	Producers (organisms that make their own food), decomposers (organisms that break down dead plants and animals)	Producers, food for small animals, decomposers	Decomposers	Producers	Consumers (organisms that eat other organisms for food)
Examples	Bacteria, archaea (a group of very old prokaryotes that can live in extreme places), cyanobacteria (blue-green bacteria that make food by photosynthesis), *Lactobacillus*, *Rhizobium*	*Amoeba*, *Paramecium*, *Euglena*, *Chlamydomonas*	Mushroom, yeast, *Aspergillus*, *Penicillium*	Moss, fern, pine, rose, grass	Ant, snake, frog, bird, cat, human

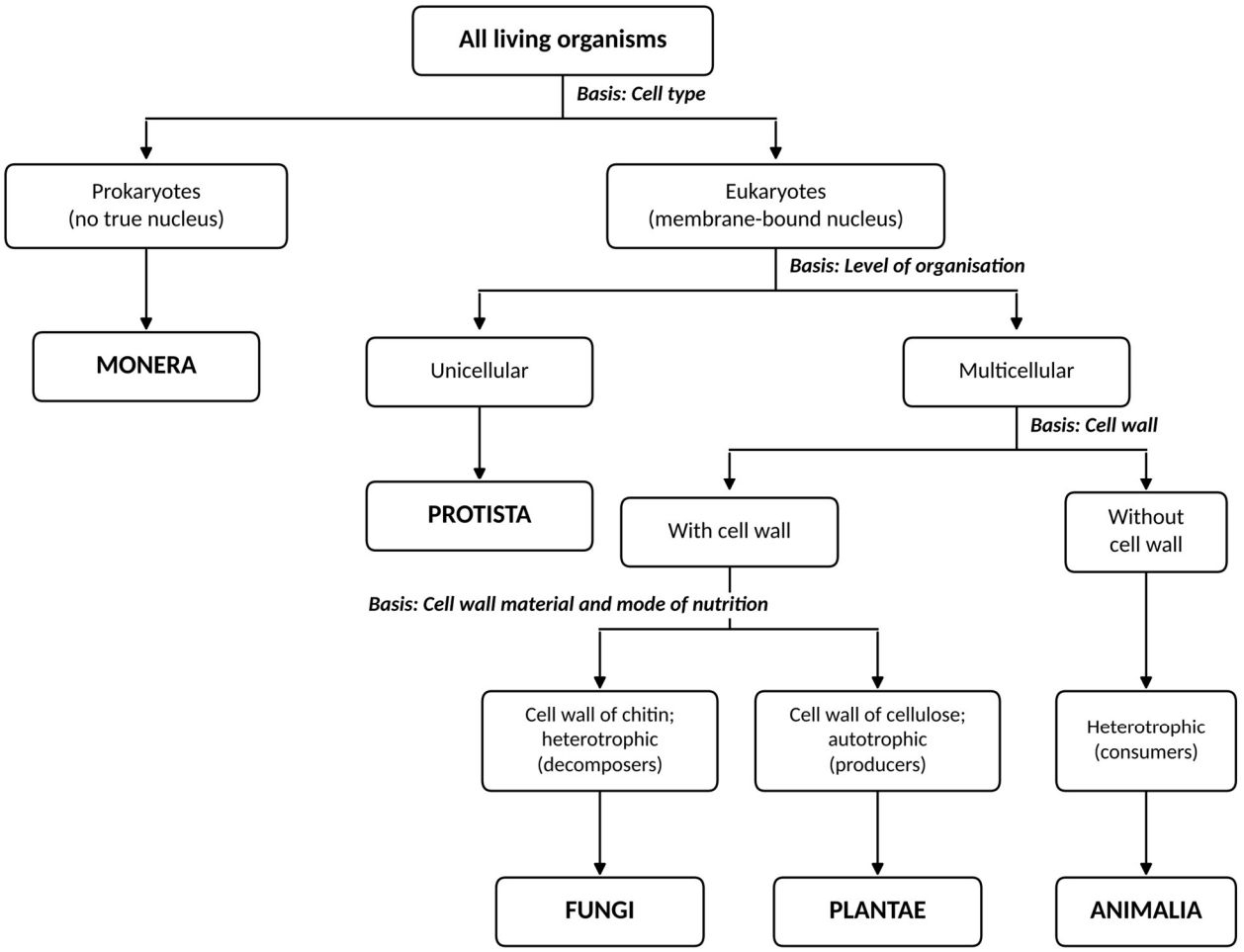


Fig. 4: Five kingdom classification shown step by step

✓ **Kingdom Monera: Unicellular prokaryotes:**

- **Main points about Monera:**
  - Bacteria are found **everywhere**: in soil, water, air, hot springs and other **extreme environments** where most organisms cannot survive. They are present even inside the human body.
  - They are found in the **gut of ruminants** (cud-chewing animals like cows) and are responsible for the production of **biogas** (fuel gas made from dung and waste) from their dung (animal waste, gobar).
  - Some bacteria are harmful and cause diseases. These are called **pathogens**. Many bacteria are useful, like ***Lactobacillus*** (turns milk into curd) and ***Rhizobium*** (fixes nitrogen in roots of pulse plants).
  - **Cyanobacteria** are **autotrophs** (make their own food) and decomposers.
  - Besides **nutrient cycling** (return of nutrients to the soil so they can be used again), some bacteria also break down **pollutants** (harmful substances that make air, water or soil dirty) like oil, pesticides (chemicals used to kill pests) and sewage (dirty waste water from homes and toilets).

  ⇒ **Know these species:**
  - ***Lactobacillus*****:** A rod-shaped bacterium that turns milk into curd by making lactic acid (a sour acid that gives curd its taste). It also lives in our gut and helps in digestion.
  - ***Rhizobium*****:** A bacterium that lives in small swellings (root nodules) on roots of pulse plants like pea and gram. It takes nitrogen from the air and makes it usable for plants, like a natural fertiliser.
  - **Cyanobacteria:** Blue-green prokaryotes that make food by photosynthesis (the process by which green plants make food using sunlight, water and carbon dioxide), for example *Nostoc* and *Anabaena*. Some of them add nitrogen to the soil of rice fields.

  ⇒ **Ram Bux Singh: Father of Modern Biogas:**
  - **Ram Bux Singh** was a pioneering (first of his kind) Indian scientist who developed biogas technology. He is popularly known as the 'Father of Modern Biogas'.
  - In **1957** he set up India's first scientifically planned biogas plant at **Ramnagar, Sitapur, Uttar Pradesh**. This was the start of modern biogas research in India.
  - He developed low-cost, efficient biogas plants for rural areas. He also served as an international consultant (an expert who gives advice) and helped set up biogas plants in several countries.
  - His work helped in renewable energy (energy from sources that do not get used up, like sunlight, wind and biogas), waste management and sustainable development.

  ⇒ **Cyanobacteria and stromatolites:**
  - Cyanobacteria (**blue-green algae**) were among the **first organisms to produce oxygen** through photosynthesis.
  - About **2.5 billion years ago**, oxygen accumulated in the atmosphere (the layer of air around the Earth). This made the Earth suitable for other forms of life.
  - Fossils (preserved remains of plants and animals) of ancient cyanobacteria are found in structures called **stromatolites** (layered rocks formed by cyanobacteria long ago). They were found in **Rajasthan** and **Madhya Pradesh**. They provide some of the earliest evidence of life on the Earth.

✓ **Kingdom Protista: Unicellular eukaryotes:**

- All **single-celled eukaryotes** without a cell wall, or with a cell wall made of **cellulose**, are grouped under the kingdom Protista.
- They are **microscopic** and highly diverse organisms.

Conceptual Learning

- **Main points about Protista:**
  - Protista includes **single-celled eukaryotic organisms** that live in **water or moist places**.
  - Some are **autotrophic** (like *Euglena* and *Chlamydomonas*). Others are **heterotrophic** (like *Amoeba* and *Paramecium*).
  - Protists are an important link in **aquatic** (of water) **food chains**. Some produce oxygen, while others serve as food for small animals.
  - Some protists act as **decomposers** and help in **nutrient cycling**.

  ⇒ **Know these species:**
  - ***Amoeba*:** A unicellular organism of pond water with no fixed shape. It moves and catches its food with finger-like **pseudopodia** (false feet).
  - ***Paramecium*:** A slipper-shaped unicellular organism covered with tiny hair-like **cilia** (tiny hair-like structures). The cilia help it swim and push food into its mouth groove.
  - ***Euglena*:** A green, spindle-shaped (thick in the middle and thin at both ends) unicellular organism with a whip-like (long and thin like a whip) **flagellum** (a long whip-like structure used for movement). It is autotrophic in light, but heterotrophic in the dark.
  - ***Chlamydomonas*:** A round, unicellular green alga (a simple green plant, mostly living in water) with two flagella (long whip-like structures used for movement) and a cup-shaped chloroplast. It lives in fresh water and is autotrophic.

✓ **Kingdom Fungi: Multicellular, heterotrophic eukaryotes with a cell wall:**

- Fungi are mostly **multicellular eukaryotes** with cell walls made of **chitin**.
- They do not make their own food. They **absorb nutrients** from dead or decaying matter through fine **filaments** (thin threads). These filaments together form a network called **mycelium**.
- Most fungi are **saprophytes** (feed on dead organic matter such as fallen leaves, twigs or dead organisms). So they play a very important **ecological role** as **decomposers**.
- They break down complex organic matter (material that comes from dead plants and animals) into simpler substances and make them readily available in the soil as **minerals** (nutrients in the soil that plants need).
- Some fungi form a **mutualistic (symbiotic)** relationship with other organisms, in which both benefit. Others live as **parasites** (organisms that live on or inside another organism and take food from it) and cause diseases in plants and animals.
- They reproduce both **sexually and asexually**, often by forming **spores** (tiny reproductive cells that can grow into a new organism). They grow best in **warm and moist** conditions.



(a) Yeast



(b) Aspergillus



(c) Mushrooms

Fig. 5: Some common fungi

- **Yeast** and **Aspergillus** mould (fungus that grows as a fuzzy layer on food) are common fungi.
- **Yeast is a unicellular organism**, but it is put under Fungi because its **cell wall is made of chitin**.
- **Mushrooms** are **macroscopic** fungi (big enough to be seen with the naked eye). They reproduce by spores.

Conceptual Learning

- *Aspergillus* and *Penicillium* are used to make **enzymes** (substances in living things that speed up chemical reactions) **and antibiotics** (medicines that kill bacteria).
- Fungi are important decomposers and help recycle nutrients. Without them, the decay of dead plants and animals would be greatly reduced. This would harm soil fertility and **ecological balance** (natural balance between organisms and their surroundings).

⇒ **Know these species:**

- **Yeast:** A unicellular fungus that reproduces by budding (a small bud grows out of the cell). It changes sugar into alcohol and carbon dioxide, so it is used to make bread soft and fluffy.
- ***Aspergillus*:** A mould seen as a green, yellow or black fuzzy (soft and hairy) layer on bread, fruits and stored grains. Some kinds are used to make enzymes and citric acid (the sour acid found in lemons), but some spoil food and make poisons.
- ***Penicillium*:** A blue-green mould often seen on old oranges and bread. The first antibiotic, penicillin, was made from it (discovered by Alexander Fleming in 1928).
- **Mushroom:** A fungus with an umbrella-shaped body above the ground and a thread network (mycelium) below. Some, like the button mushroom, are eaten, but some are poisonous.

⇒ **Wild edible mushrooms:**

- **Wild edible mushrooms** are a valuable food with high nutritional and medicinal value.
- They are important forest resources used and conserved by many communities, including tribal communities in India. These communities have traditional knowledge about edible and poisonous mushrooms based on **folk taxonomy** (traditional local system of naming and grouping).
- Some wild mushrooms are used as food and some as medicine for treating various **ailments** (illnesses).
- Mushroom cultivation is becoming a promising **vocation** (occupation) for livelihood. It needs minimal space and low investment, and has a fast life cycle (**30 to 45 days**).

✓ **Kingdom Plantae: Multicellular, autotrophic eukaryotes with a cell wall:**

- Plants are **multicellular, autotrophic** organisms that perform **photosynthesis**.
- Their cells have a **rigid** (hard and not bending) **cell wall** made mainly of **cellulose**, which gives support and protection.
- Plants form the base of most food chains and release **oxygen**, which is essential for life on the Earth.
- Kingdom Plantae is further divided into five classes: **Thallophyta, Bryophyta, Pteridophyta, Gymnosperm and Angiosperm**.

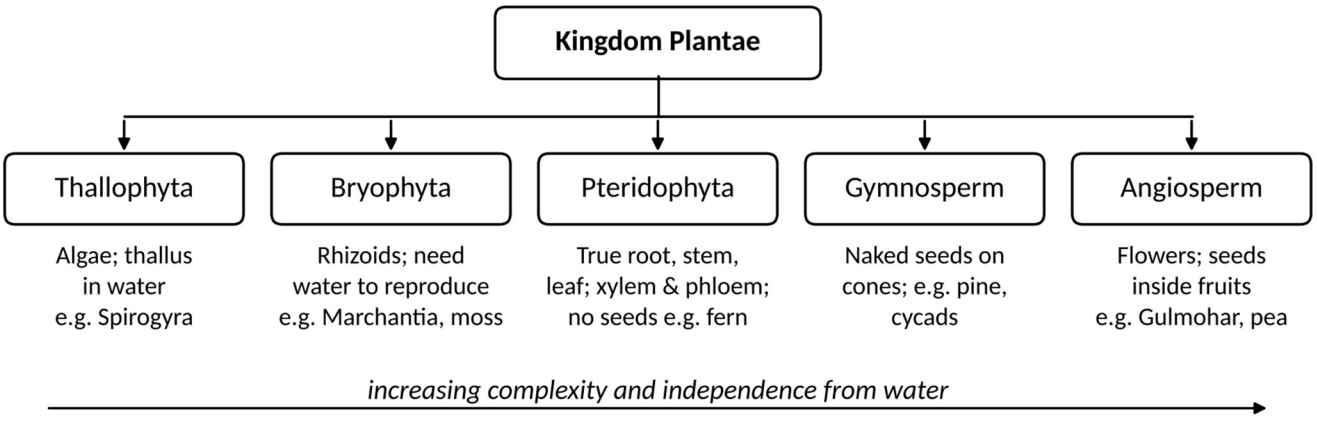


Fig. 6: The five classes of Kingdom Plantae

Conceptual Learning

- **(a) Thallophyta (algae): Primitive plants:**
  - *Thallos* means **undifferentiated body** (body not divided into root, stem and leaves); *phyton* means plant.
  - Thallophytes are among the simplest plant forms. They are mostly found in **water or moist environments**.
  - Their body is a **thallus**, a simple body that allows **direct exchange** of gases, nutrients and water with the surroundings.
  - They are well adapted (suited to live in a particular place) to **aquatic habitats**. Example: ***Spirogyra***.

  ⇒ **Know these species:**
  - ***Spirogyra*****:** A green alga seen as slimy, hair-like green threads in ponds, also called 'pond silk'. Its name comes from its spiral, ribbon-shaped chloroplasts (green parts of the cell where photosynthesis takes place).

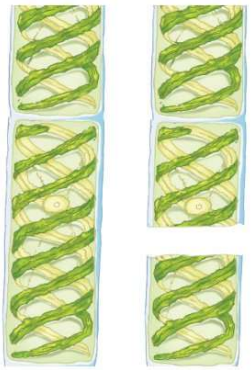


Fig. 7: Spirogyra: filamentous (thread-like) alga with spiral, ribbon-shaped chloroplasts

  ⇒ **Lichens:**
  - The white-green patches seen on tree trunks, damp walls and stones are **lichens**. Elderly people in villages recognise these patches as an indicator (something that shows a condition) of a **pollution-free environment**.
  - Lichens change their colour with air pollutants. From their colour, researchers can find the concentration of pollutants in the air. So lichens are natural **bioindicators** (living indicators) for measuring air quality.
  - Villagers collect lichens as forest produce. One kind, called *patthar ke phool*, is used as a **spice**. Some are used as **medicine**, and since old times some are used to make **dyes** (colours used for colouring cloth) that colour wool and silk maroon (dark brownish-red colour), violet or burgundy (dark purplish-red colour).
  - Biologically, lichens are **symbiotic** (mutual association of two organisms): one partner is an **autotrophic alga** and the other is a **heterotrophic fungus**. The fungal partner provides protection and the algal partner photosynthesises and provides food.
  - Some lichens are poisonous. So they must be identified and classified correctly before use.

- **(b) Bryophyta: First steps on land, still need water:**
  - *Bryon* means moss; *phyton* means plant. Examples: mosses and liverworts (small, flat, green plants related to mosses) like ***Marchantia*** and **moss**.
  - They represent an important **shift from water to land**.
  - We see them as green mats on damp rocks, walls of old buildings and soil surfaces, especially during the monsoons (rainy season). They survive in **moist and shady** places.

Conceptual Learning

- They have a **more differentiated body** than thallophytes. They have root-like structures called **rhizoids**, and may have simple stem-like and leaf-like structures. These are not true roots, stems and leaves.
- They lack **vascular tissues** (xylem and phloem) for transporting water and food.
- They require water for reproduction, as the male reproductive cells must swim to reach the female cells. That is why they are called the **'amphibians of the plant kingdom'** (like amphibians, they live on land but need water to reproduce).
- They show how plants began to **colonise** (settle on) land, while remaining dependent on moisture.

⇒ **Know these species:**

- ***Marchantia*:** A liverwort with a flat, green, ribbon-like body that lies on wet soil. It is fixed to the soil by rhizoids and has small cups on top that help in reproduction.
- **Moss:** Small, soft plants that grow close together like a green carpet on wet walls and rocks. Thin stalks with small capsules (small box-like parts that hold spores) on top carry spores.



(a) Marchantia



(b) Moss

Fig. 8: Members of class Bryophyta

- Bryophytes stay small and need moist places because they lack vascular tissues (xylem and phloem) and must absorb water through their whole body surface.

(c) Pteridophyta: Adaptation to land and having structural transport system:

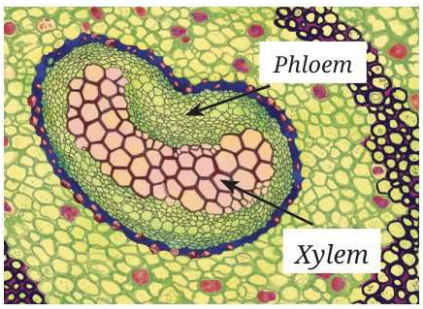
- *Pteris* comes from *pteron*, meaning feather; *phyton* means plant. Example: **ferns**.
- They have **true roots, stems and leaves**.
- They have **specialised transport tissue** called **vascular tissues**. **Xylem** transports water and **phloem** transports food throughout the plant.
- To test this, a cross section (a thin slice cut across) of a fern stem is cut (or a permanent slide is used) and seen under a microscope. The xylem and phloem can be seen clearly.
- They require **aquatic conditions for reproduction** and do **not produce seeds**.

⇒ **Know these species:**

- **Fern:** A plant with feather-like leaves called fronds (large divided leaves of ferns). Young leaves are rolled up like a coil, and brown dots under the leaves hold spores.



(a) Fern



(b) Cross section of fern stem

Fig. 9: Fern plant and cross section of its stem (xylem in the middle, phloem around it)

- **(d) Gymnosperms: Reproduction without water:**
  - *Gymnos* means naked; *spermos* means seed. Examples: **pines** and **cycads**.
  - They are well adapted to **cold and dry regions**. Their **needle-like or scale-like leaves** reduce water loss and withstand harsh conditions.
  - They produce **seeds**, which protect the developing **embryo** (young plant inside the seed) and contain stored food.
  - They do **not require aquatic conditions for fertilisation** (joining of male and female reproductive cells).
  - Their seeds are **not enclosed in fruits** (naked seeds) and are often exposed on **cones** (woody structures on pine trees that carry seeds).

  ⇒ **Know these species:**
  - **Pine:** A tall evergreen (that stays green throughout the year) tree of cold hilly areas like the Himalayas, with needle-like leaves and woody cones. It gives us wood and resin (used to make turpentine).
  - **Cycads (e.g. *Cycas*):** Palm-like plants with a thick stem and a crown of stiff, hard leaves at the top. They are among the oldest seed plants and are often grown in gardens.



Fig. 10: Pine cone with naked seeds

- **(e) Angiosperms: Efficient reproduction and seed dispersal:**
  - *Angeion* means vessel (container); *spermos* means seeds. They are also called **flowering plants**. They have the **most complex body organisation** among plants.
  - They produce **flowers and fruits**. Flowers attract **pollinators** (animals like bees, birds and bats that carry pollen from flower to flower), increasing the efficiency of reproduction. Fruits help in **seed dispersal** (spreading seeds to new locations).
  - These features allow angiosperms to occupy a wide range of environments. So they are the **most diverse plant group** on the Earth. Example: **Gulmohar**.

  ⇒ **Know these species:**
  - **Gulmohar:** A shady tree that is covered with bright red-orange flowers in summer. Its seeds are kept inside long, flat, brown pods (fruits).

  ⇒ **Hortus Malabaricus: an early book on Indian plants:**
  - *Hortus Malabaricus* is one of the earliest scientific books on Indian plants. It was compiled (collected and written) in the **17th century** by **Hendrik van Rheede** with the help of **Itty Achudan** (an Indian herbalist, botanist and physician) and other local experts.
  - It describes hundreds of plant species and their medicinal uses. It shows that traditional knowledge and science can work together.

Concept Learning



Fig. 11: An angiosperm tree: Gulmohar in flower

- **Overall pattern in plants:**
  - From algae to angiosperms, plant groups show a sequence of **structural changes** that help plants meet the challenges of life on land.
  - Early plants relied on water for support and reproduction. Gradually, plant groups **evolved** transport tissues, seeds and flowers.
  - The diversity of plant forms reflects many ways of balancing growth, transport, reproduction and survival in different environments.

✓ **Kingdom Animalia: Multicellular, heterotrophic eukaryotes:**

- Animals are **multicellular** and **heterotrophic** organisms. They depend on other organisms for their food. Their cells have no cell wall.

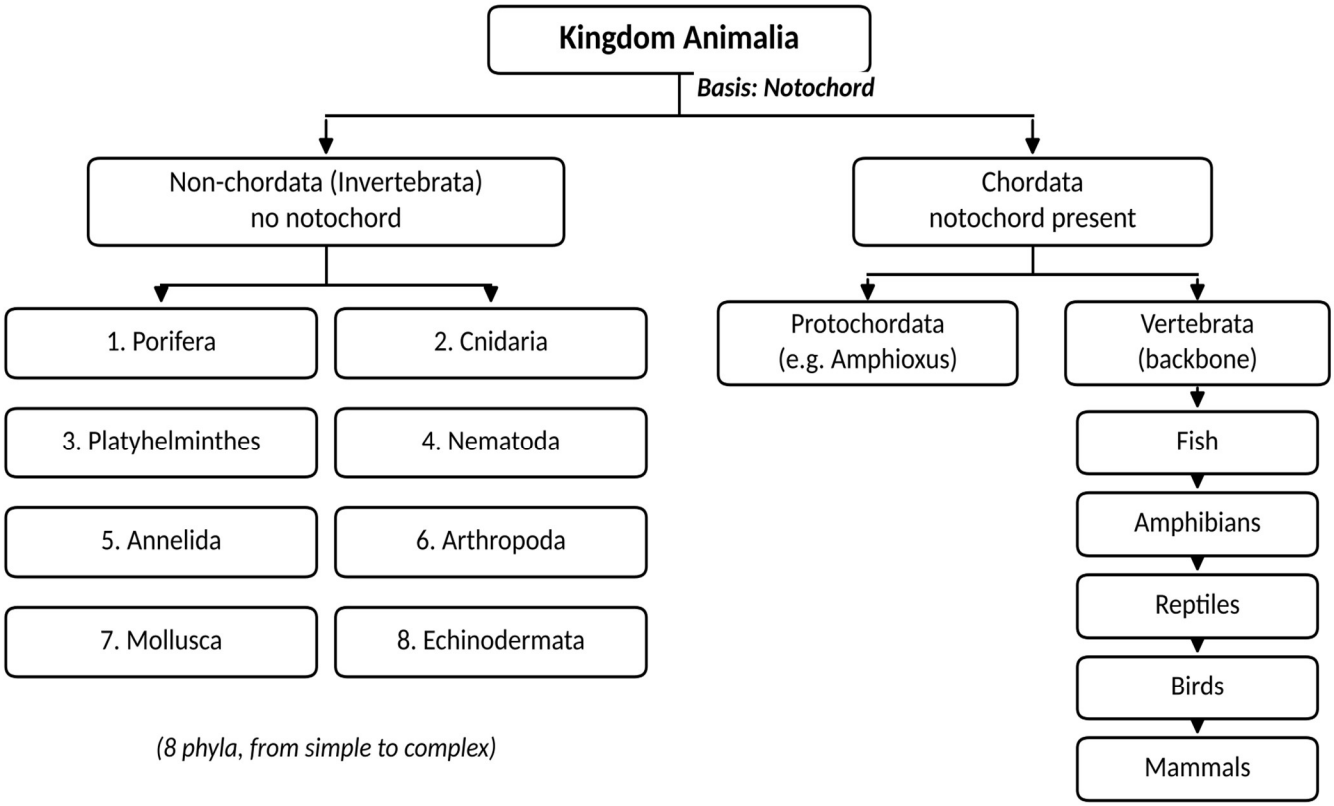


Fig. 12: Classification of Kingdom Animalia

- Most animals show **locomotion** (movement from one place to another), **rapid response to stimuli** (changes around them) and **coordinated behaviour**. These help them actively search for food, avoid **predators** (animals that hunt them) and interact with their surroundings.
- One of the major criteria for classifying animals is the presence or absence of a **notochord**. It is a **flexible, rod-shaped** structure.
- Animals without a notochord are **Non-chordata (Invertebrata)**. Animals with a notochord are **Chordata**.
- In some chordates (animals that have a notochord), the notochord acts as a **precursor** (the structure that comes before) for the development of the **vertebral column** (backbone).
- Chordata is further divided into **Protochordata** and **Vertebrata**.
- **Invertebrates: Animals without a notochord:**
  - Invertebrates (animals without a backbone) lack a notochord, yet they show a wide range of **body organisation**, from a simple structure to complex organ systems (a group of organs working together).
  - Studying them shows how the animal body gradually changed over time, leading to advancement (progress, improvement) in movement, diverse feeding habits and better control and coordination (working together of different body parts), while also introducing exceptions and new challenges.
- **(1) Porifera (pore-bearers): Multicellularity without tissues:**
  - Example: **sponges**. They represent one of the simplest animal body plans.
  - They are **multicellular** but lack organisation of **tissues and organs** (cellular level of organisation).
  - Numerous **pores** in their body allow water to flow continuously through it. This water brings food particles and oxygen directly to individual cells and carries the waste away.
  - They remain **fixed in one place** and are found in **aquatic environments** (mostly marine).
  - **Sponges cannot survive on land**, because they cannot move to find food, and without water flowing through the pores they get no food or oxygen, waste is not removed, and the body dries up.

  ⇒ **Know these species:**
    - **Sponge:** A simple animal that stays fixed to rocks, mostly in the sea, and looks like a plant. Its body is full of holes, and dried skeletons of some sponges were once used as bath sponges.

  ⇒ **Sponges as water filters:**
    - Sponges filter a huge amount of water. One kilogram of sponge can filter up to **24,000 litres** of sea water in a day.
- **(2) Cnidaria: True tissues and active feeding:**
  - Examples: *Hydra*, jellyfish and corals.
  - They show **tissue-level organisation**, which allows **specialised cells** to perform specific functions, such as **tentacles** (long, thin, arm-like parts around the mouth) used for capturing prey (animal that is hunted and eaten by another animal).
  - Unlike poriferans (members of Porifera, that is, sponges), they catch their prey using tentacles instead of depending on water currents to bring food particles inside through pores.
  - They have a **single opening** that works for both food intake and waste removal.
  - **Limitation of a single opening:** the animal cannot take in food and remove waste at the same time. New food may mix with waste. Digestion cannot happen step by step in different parts. This limits the size and working of the body.

  ⇒ **Know these species:**
    - ***Hydra*:** A tiny freshwater animal, about 1 cm long, that sticks to water plants. It catches small animals with stinging tentacles and can regrow lost body parts.

Conceptual Learning

- **Jellyfish:** A sea animal with a soft, umbrella-shaped, see-through body and long hanging tentacles. Its body is mostly water and its sting can hurt.
- **Corals:** Tiny cnidarians that live together in large groups and make hard skeletons of calcium carbonate (a hard white substance; chalk and limestone are made of it). Over many years these form coral reefs, like those near the Andaman and Lakshadweep islands.

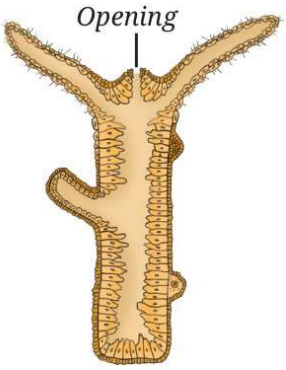


Fig. 13: Longitudinal section of Hydra showing tissue level organisation and its single opening

- **(3) Platyhelminthes (flatworms): Bilateral symmetry and directional movement:**
  - They show **bilateral symmetry** (the body can be divided into two equal halves along **one plane** only). The body has distinct **head-tail** and **front-back** regions.
  - This allows better coordination of movement. It is the **beginning of directional activity**.
  - Their **flattened bodies** permit efficient **diffusion** (movement of substances from where they are more to where they are less) **of gases** without specialised **respiratory organs** (organ level of organisation).
  - There is still a **single opening** for food intake and waste elimination.
  - Many flatworms have a **parasitic lifestyle**. Parasitic forms have **hooks and suckers** to attach firmly to the **host** tissues for obtaining nutrients.

  ⇒ **Know these species:**
  - **Flatworms:** *Planaria* is a free-living flatworm of fresh water that can grow a whole new body from a small piece. Tapeworm (a long, flat, ribbon-like worm that lives in the intestine) and liver fluke (a flatworm that lives in the liver) are parasitic flatworms that live inside other animals, including humans.

  ⇒ **Parasitic worms and our health:**
  - Some **infectious** (that can spread from one person to another) **diseases** are caused by parasitic worms. They enter our body through **contaminated** (dirty) water and food.
  - They live inside our body, especially in the **alimentary canal** (digestive system), and take nutrients from our body. The organism in or on which a parasite lives is called the **host**.
  - So we should wash hands properly, maintain personal hygiene (cleanliness) and good sanitary (related to toilets and cleanliness) habits, eat properly cooked food and drink boiled or filtered water.
- **(4) Nematoda (roundworms): Efficient body design with two openings:**
  - They have **elongated** (long and stretched)**, cylindrical** (shaped like a pipe or tube) bodies. This body structure allows efficient movement through soil, water or **host tissues**.
  - This increases the range of environments in which they can survive, despite a relatively simple internal organisation.

Conceptual Learning

- Their body has **two openings: mouth and anus** (the opening through which undigested food leaves the body).
- They show **organ system level** of body organisation, and **male and female worms are distinct** (separate).

⇒ **Know these species:**

  - **Roundworms:** *Ascaris* lives in the human intestine (the long tube in the body where food is digested and absorbed) and spreads through dirty food and water. Another roundworm, the filaria worm, causes elephantiasis (swelling of legs).

- Two openings make digestion better, as food moves in one direction only and food intake and waste removal can happen at the same time.

- **(5) Annelida (segmented worms): Segmentation and body cavities:**
  - Example: **earthworms**. Their bodies are **cylindrical** and divided into **segments** (ring-like parts).
  - They have **organ system level** of organisation. **Muscles** help them in **locomotion**, and a **nerve cord** (a long bundle of nerves that carries messages in the body) helps in control and coordination.
  - **Segmentation** allows greater flexibility and more precise control of movements.
  - A **body cavity** (space inside the body) is present.

  ⇒ **Know these species:**

    - **Earthworm:** It lives in wet soil and eats soil and dead leaves. It makes the soil loose and fertile, so it is called the 'farmer's friend'.

- **Advantages of segmentation (seen from the way an earthworm moves):**

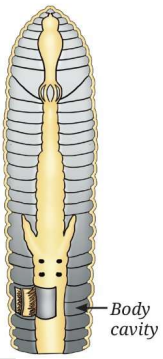


Fig. 14: Internal structure of an earthworm showing organ formation and body cavity

  - Each segment can contract or relax on its own, so the worm moves in waves, bends easily and burrows into soil.
  - If one segment is injured, the rest of the body still works.
  - Segments can become specialised for different functions (as in arthropods).

- **(6) Arthropoda: Jointed appendages and an external skeleton:**
  - *Arthro* means limbs; *poda* means **appendages** (jointed legs and other parts attached to the body). Examples: insects, crabs and spiders.
  - They have **segmented bodies**, with different segments specialised for different functions.
  - Their defining feature is a hard **external skeleton** (exoskeleton: a rigid external covering).
  - **Advantages of the outer covering:** it provides protection, **reduces water loss** and supports powerful muscles. So arthropods can survive in dry and exposed environments.

  ⇒ **Know these species:**

Conceptual Learning

- **Insects:** They have 6 jointed legs and a body in 3 parts (head, thorax, abdomen), for example cockroach, butterfly and ant. They are the largest group of animals on the Earth.
- **Crab:** It has a hard shell and 5 pairs of jointed legs; the front pair are big claws. It lives in the sea, in rivers and on sea shores.
- **Spider:** It has 8 jointed legs, so it is not an insect. It spins silk webs to catch insects for food.

- **(7) Mollusca: Organ system level organisation with soft bodies:**
  - Examples: **snails, squids and octopuses**. They show **organ system level** organisation with **soft bodies**.
  - In many molluscs, a **shell** develops and protects the soft body.
  - The body has a distinct **head**, a **muscular foot** (a strong, fleshy part used for moving) and a **hump** (raised part on the back that holds the main body organs).
  - This group shows how a basic **body plan** can be modified in different directions depending on environmental demands.

  ⇒ **Know these species:**
  - **Snail:** It has a soft body inside a coiled shell and moves slowly on its muscular foot, leaving a slimy trail. It is found on wet land and in water.
  - **Squid:** A fast-swimming sea animal with a long body and ten arms; its shell is small and inside the body. It squirts dark ink to escape enemies.
  - **Octopus:** A sea animal with a soft, round body and eight arms with suckers, and no shell. It is very clever and can change its colour to hide.

- **(8) Echinodermata: Internal support without a notochord:**
  - *Echinos* means spiny; *derma* means skin. Examples: **starfish and sea urchins**. They are **marine** (live in the sea).
  - They have a hard **internal skeleton** (endoskeleton) made of **calcium carbonate**, although they lack a notochord.
  - This internal skeleton provides protection and controlled movement.
  - Their body organisation is quite like that of more complex animals, showing a gradual shift towards **internal skeletal support**.

  ⇒ **Know these species:**
  - **Starfish:** A star-shaped sea animal, usually with 5 arms. It moves with many tiny tube feet (tiny tube-like feet used for moving) and can regrow a lost arm.
  - **Sea urchin:** A round sea animal covered with long, sharp spines. It lives on the sea floor and feeds on algae.

Table D: The animal kingdom: invertebrate phyla compared

Phylum	Example	Habitat	Level of organisation	Skeleton
Porifera	 *Sponge*	Water (marine)	Cellular	None
Cnidaria	 *Jellyfish*	Water (fresh and marine)	Tissue	None

Conceptual Learning

Phylum	Example	Habitat	Level of organisation	Skeleton
Platyhelminthes	 *Flatworm*	Water / inside host	Organ	None
Nematoda	 *Roundworm*	Soil / water / inside host	Organ system (digestive system)	None
Annelida	 *Earthworm*	Moist soil / water	Organ system	None
Arthropoda	 *Crab*	Land / water	Organ system	Exoskeleton
Mollusca	 *Snail*	Water / moist land	Organ system	Exoskeleton (shell)
Echinodermata	 *Starfish*	Marine water	Organ system	Endoskeleton

Fig. 15: The animal kingdom: invertebrate phyla with example animals

- **Looking across invertebrates:**
  - From sponges to echinoderms, animal body structure shows a pattern of **increasing complexity in body organisation**: cellular, then tissue, then organ, then organ system level.
  - New structural features improve feeding, movement and protection, but often introduce new challenges.
  - These help explain survival in diverse environments.
- **Protochordates: The appearance of the notochord:**
  - Example: ***Amphioxus***. They have a **notochord at least once in their life**.
  - The notochord provides **internal support without restricting movement**. This was a crucial (very important) change in body organisation.
  - Protochordates are a **primitive** (very early and simple) **type of chordates**. They help us understand how animals with a notochord may have arisen from simpler forms (the link between invertebrates and vertebrates).

  ⇒ **Know these species:**

  - ***Amphioxus*:** A small, see-through, fish-like sea animal that lives buried in sand. It keeps its notochord for its whole life.

Conceptual Learning

- **Vertebrates: Animals with a backbone:**
  - Vertebrates possess a **vertebral column** (backbone). It is an internal skeletal structure that supports the body and protects **vital organs** (organs needed to stay alive) like the **brain and the spinal cord** (a long bundle of nerves running inside the backbone).
  - This internal framework allows a **larger body size**, efficient movement and complex organ systems. Vertebrates show **advanced sensory abilities** (abilities related to senses like seeing, hearing and smelling) and coordinated behaviour.
  - They are classified into five groups: **fish, amphibians** (animals that live both on land and in water, like frogs)**, reptiles** (crawling animals with scales, like snakes and lizards)**, birds and mammals** (animals that feed milk to their young ones), on the basis of their **habitat use** (the way organisms use the places they live in), **body covering** and **reproduction**.

  ⇒ **Forests with rich biodiversity as a barrier against disasters:**
    - Forests with rich and highly diverse **flora** (plants) and **fauna** (animals) create physical, biological and chemical **barriers** and reduce the impact of disasters.
    - **Physical barrier:** the large diversity of trees in **mangrove** (trees that grow in salty water along sea coasts) **forests** helped during the **super cyclone** (a very strong storm with heavy rain and fast winds) **that hit Orissa in 1999**. Villages with more mangroves had less destruction.
    - **Biological barrier:** rich biodiversity generally means lower risk of **tick-borne** (spread by ticks; ticks are tiny blood-sucking creatures, related to spiders, that stick to the skin of animals) **diseases**. In the Western Ghats, forest diversity acts as a barrier against **Monkey Fever (Kyasanur Forest Disease, KFD)**, because many animals are hosts in which the virus cannot **replicate** (multiply).
    - **Chemical barrier:** diverse **microorganisms** (organisms that can be seen only with a microscope) in forest soils and plant roots absorb, transform or break down pollutants. This improves water quality. Mangrove soils trap **sediments** (mud and sand) and heavy metals (harmful metals like lead and mercury) and stop pollution from spreading into oceans and rivers.

Adaptations as Outcomes of Structural Change

➢ The diversity seen in animals today is an **outcome of changes in body structure** over long periods of time.

➢ **Fins and gills** (breathing organs of fish and tadpoles) allow fish to move and breathe in water.

➢ **Feathers and hollow bones** (bones that are empty inside) make flight possible in birds (hollow bones make the body light).

➢ **Fat storage in camels** and **thick fur in polar bears** show how structural features support survival under **extreme conditions**.

➢ **Mammary glands** in mammals (glands that make milk) improve the survival of young ones.

➢ These features are **interconnected**. They show how different vertebrates tackle the challenges of survival in different environments.

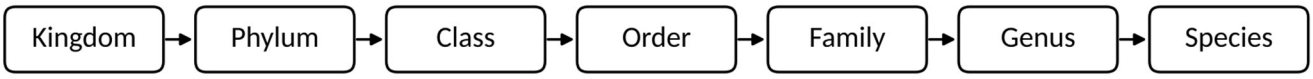
➢ Animal diversity reflects a wide range of body forms found in organisms living under different environmental conditions.

✓ **The hierarchical nature of classification:**

- **Hierarchy** means arrangement in levels, one above the other.
- Classification follows a **step-by-step order**, starting from very **broad groups** and moving towards smaller and **more specific** ones.
- At each lower level, organisms share **more common features**. Every lower group is a part of the group above it.

Conceptual Learning

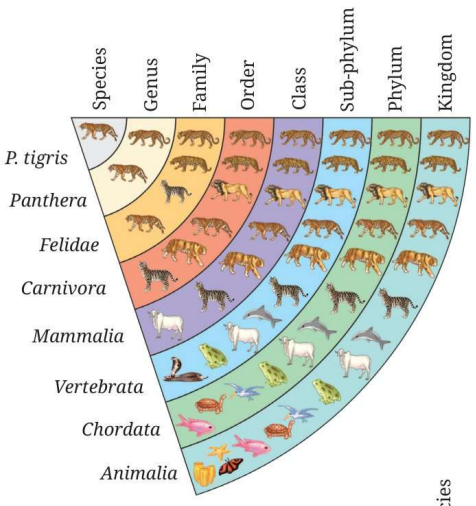
- Levels: **Kingdom → Phylum** (a large group within a kingdom) **→ Class → Order → Family → Genus → Species**.
- It works like an **address**. A house address helps us locate a place exactly. In the same way, classification helps scientists identify, compare and study organisms accurately, and understand how they are related.



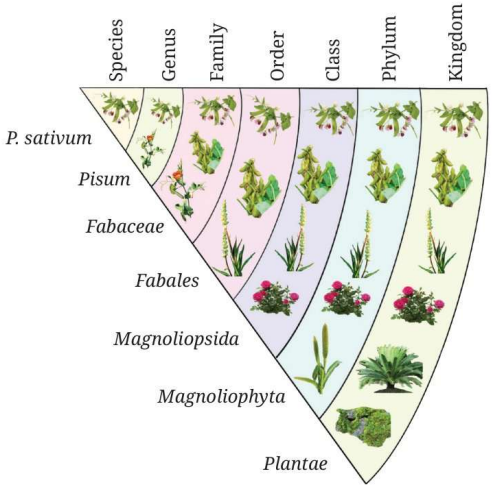
broadest group, most members → narrowest group, most features in common

Fig. 16: Levels of classification from kingdom to species

Level	Tiger	Pea
Kingdom	Animalia	Plantae
Phylum	Chordata	Magnoliophyta
Sub-phylum	Vertebrata	-
Class	Mammalia	Magnoliopsida
Order	Carnivora	Fabales
Family	Felidae	Fabaceae
Genus	*Panthera*	*Pisum*
Species	*P. tigris*	*P. sativum*



Tiger	
Kingdom	*Animalia*
Phylum	*Chordata*
Sub-phylum	*Vertebrata*
Class	*Mammalia*
Order	*Carnivora*
Family	*Felidae*
Genus	*Panthera*
Species	*P. tigris*



Pea	
Kingdom	*Plantae*
Phylum	*Magnoliophyta*
Class	*Magnoliopsida*
Order	*Fabales*
Family	*Fabaceae*
Genus	*Pisum*
Species	*P. sativum*

Fig. 17: Pyramids showing levels, from kingdom at the top to species at the base, with the examples of a tiger and a pea plant

Scientific Naming: The Binomial System

- A tiger is called *bagh* in Hindi, *puli* in Tamil, tiger in English and *tigre* in French. If people from different regions discuss this animal, confusion may arise.
- To avoid this, scientists use a **universal system** of naming living organisms called **binomial nomenclature** (binomial: two names; nomenclature: system of naming).
- This system was introduced by **Carolus Linnaeus** in the 18th century.
- According to this, every organism has a **scientific name** with **two parts**, written in **Latin** or a **Latinised** (changed into Latin form) form.

S. No.	Common name	Scientific Name
1.	Tiger	*Panthera tigris*
2.	Mango	*Mangifera indica*
3.	Lion	*Panthera leo*
4.	Pea	*Pisum sativum*
5.	Edible camphor	*Cinnamomum camphora*
6.	Purple frog	*Nasikabatrachus sahyadrensis*

- The first word (*Panthera*, *Mangifera*) is the **genus**. The second word (*tigris*, *indica*) is the name of the **species**.
- A **genus** groups closely related species that share common features. Example: *Panthera tigris* (tiger) and *Panthera leo* (lion) are both under the genus *Panthera*, the **roaring cats** (able to roar), with similar skull structure.
- A **species** is a group of organisms with similar individuals capable of **interbreeding** (breeding among themselves) and producing **offspring** (young ones).
- Together, the genus and species names form a **unique scientific name used worldwide**.
- ⇒ **Know these species:**
  - **Tiger (*Panthera tigris*):** The national animal of India and the biggest member of the cat family, with a striped orange coat. It lives alone and is protected under Project Tiger (started in 1973).
  - **Lion (*Panthera leo*):** In India, wild lions (Asiatic lions) live only in the Gir forest of Gujarat. Lions live in groups called prides (groups of lions), and males have a mane.
  - **Mango (*Mangifera indica*):** A large tree native (originally belonging to that place) to India; its fruit is the national fruit of India. Each mango fruit has one big seed inside.
  - **Pea (*Pisum sativum*):** A climbing plant whose seeds grow inside a pod (fruit) and are eaten as a vegetable. Its roots have nodules with *Rhizobium*, and Mendel (a scientist known as the father of genetics) used it in his experiments.
  - **Edible camphor (*Cinnamomum camphora*):** A tree whose leaves and wood give camphor. Camphor is used in medicines for pain, swelling and cough.



Fig. 18: The two parts of a scientific name

Conceptual Learning

- **Rules for writing the scientific names:**
  1) The name has two parts: genus and species.
  2) The genus name begins with a **capital letter** and comes first. It is followed by the species name, which is written in **small letters (lower case)**.
  3) The scientific name is written in **italics** (slanting letters) when printed, or **underlined** when handwritten.

⇒ **Three domain system:**

- The five kingdom classification was more **comprehensive** (complete) than earlier systems, but it still could not fully explain the diversity of life.
- With advances in microscopes and **genetic studies**, scientists began to compare organisms at the **DNA level**. Every living cell contains genetic material (DNA), which carries the instructions for its growth and function.
- Organisms with similar DNA are considered to have a **common ancestry** (the line of earlier forms from which an organism has come).
- Based on the genetic data, **Carl Woese (1977)** proposed the **three domain system**: **Bacteria, Archaea and Eukarya**.
- This system showed that **microscopic life forms** are far more diverse than previously believed.

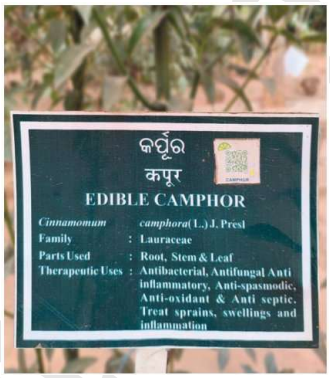


Fig. 19: A plant name board showing a scientific name: Edible camphor, Cinnamomum camphora, family Lauraceae, with parts used and medicinal uses

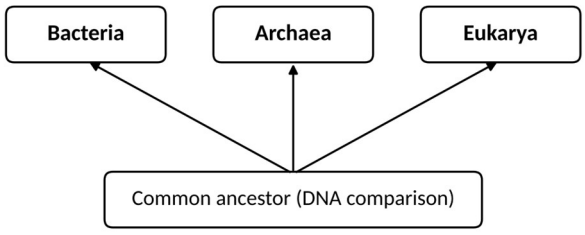


Fig. 20: The three domains of life

⇒ **Why classification systems keep changing:**

- Science evolves and progresses as we learn new things. Aristotle's system worked for his time.
- New tools (like **microscopes**) and technological advancements (like **staining techniques**) helped scientists discover **microorganisms**. This helped us understand the natural world better, and so the biological classification changed.
- So, biological classification (and science) is an **ongoing process of reasoning and change**.

Fossils as Evidence

- **Fossils** are **preserved remains** of plants and animals found in layers of rocks, sand and mud.
- Generally, **older layers contain simpler organisms**, while **newer layers show more complex forms**.
- Important fossils of giant dinosaurs, early humans and ancient plants have been discovered in India.
- Fossils act as **natural records** that help us understand how life has changed over millions of years.

⇒ **Scientist Birbal Sahni:**

- **Birbal Sahni** was an eminent (famous) Indian scientist who studied **fossil plants**.
- He founded the **Birbal Sahni Institute of Palaeosciences (BSIP), Lucknow**, which continues his work on ancient plants and past environments even today.
- His studies linked present-day plants with their **ancestors**. They showed that life on the Earth has a long, connected history.
- His work continues to inspire young scientists to explore how fossils reveal the story of our planet.

Biodiversity Under Threat

- Each species, large or small, plays an important role in nature. Plants produce food and oxygen, animals pollinate flowers and **disperse** seeds, and **microorganisms** recycle nutrients.
- Human activities like **pollution, deforestation** (cutting of forests), **overuse of resources** and **climate change** are reducing biodiversity.
- When one species disappears, others that depend on it may also **decline** in number and finally disappear.

⇒ **Purple Frog: naming a new species:**

- When scientists discover a new organism, they compare it with the already known species. Sometimes the species name reflects the place of discovery, or honours a scientist.
- The **Purple Frog** (*Nasikabatrachus sahyadrensis*) of **Kerala** is named after the **Sahyadri Hills**.
- It lives **under the ground** for most of the year and comes out only in the **monsoon to breed**.
- It belongs to an **ancient family of frogs**, as understood from fossils.
- Its discovery in **2003** helped scientists understand ancient amphibian groups and highlighted the need for **biodiversity conservation** in the Western Ghats.



Fig. 21: A purple frog resting on wet soil

Conceptual Learning



Fig. 22: Sangai deer on the phumdis (floating grasslands) of Loktak lake

⇒ **Phumdis and the Sangai deer:**

- **Phumdis** (local name) are **floating grasslands** in **Loktak lake** of the **Keibul Lamjao National Park, Manipur**. They are one of the unique habitats of the world.
- They are floating beds of soil with a variety of **rooted vegetation** growing on them, with rich organic matter. Their thickness is from a few centimetres to two metres.
- The habitat is home to an **endangered** variety of **Sangai**, the 'dancing deer', **endemic** to Manipur. The deer spends considerable time on the phumdis.
- This deer species was declared **extinct** (no longer living anywhere on the Earth) in **1951**. In **1953** it was **rediscovered** (found again) by naturalists (people who study nature, plants and animals) on the basis of its unique characteristics of having **hooves** (hard covering on the feet of animals like deer and cows) and **elongated patterns**. These characteristics helped in its identification.
- Recently, the phumdis are **degenerating** (getting damaged) and the Sangai deer is listed in the **IUCN** (International Union for Conservation of Nature, a world body that keeps the list of threatened species) **Red Data list** (list of threatened species). Habitat loss will significantly affect the Sangai population. Efforts to conserve the phumdis and the Sangai are in progress.
- **Ways to save the habitat and the Sangai:** stop pollution and silt (fine mud carried by water) from entering Loktak lake; keep the water level natural so the phumdis stay thick; stop over-cutting of phumdi plants and overfishing (catching too many fish); stop hunting; stop new settlements and grazing inside the park; breed some deer in a safe second place; involve local people and spread awareness; keep regular scientific watch.

Activities

✓ **Activity 1: Let us compare and classify:**

- **Aim:**
  - To observe the diversity of animals in an ecosystem and group the organisms in more than one way.

Conceptual Learning

- **Steps:**

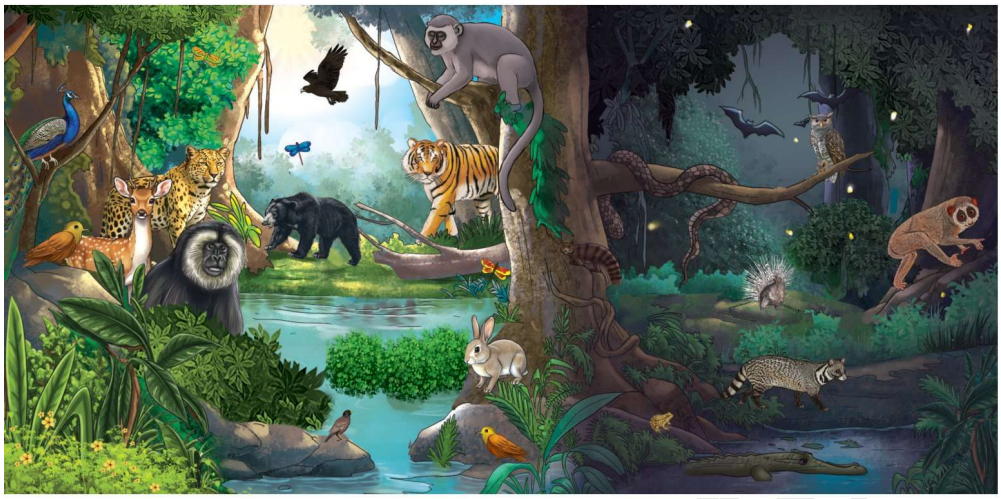


Fig. 23: Day and night view of an ecosystem

1) Observe the picture of the ecosystem (day and night view) carefully and guess how the organisms are grouped in it.
2) Identify the animals in the picture, note where they are seen (flying high in air, flying near the ground, tree, water or forest floor) and whether they seem active during the day, during the night, or both.
3) Record the observations in an observation table (Table 12.1).
4) Group the same organisms in more than one way, changing the grouping criterion each time (Table 12.2).

- **Observations:**
  - **Grouping in the picture:** the left side is bright (day) and the right side is dark (night). So the animals are grouped by **when they are active**: during the day, during the night, or both.
  - **Animals seen in the picture:** peacock, spotted deer, leopard, tiger, sloth bear (a shaggy black bear found in Indian forests), lion-tailed macaque, langur (grey monkey on the tree), eagle/kite (flying high), dragonfly, butterflies, small birds, rabbit, frog, crocodile/gharial (in water), bat, owl, python (on a branch), porcupine (an animal covered with sharp spines), civet (a small cat-like wild animal), slender loris (a small, slow-moving animal with very big eyes that lives on trees) and fireflies (insects whose body gives out light).

Table A: Observation table for recording the data (Table 12.1 filled)

Organism	Where do you see it?	When does it appear to be active?	Any visible feature(s)
Owl	Tree	Night	Feathers, big eyes facing front
Peacock	Tree / forest floor	Day	Feathers, long colourful tail
Eagle / kite	Flying high in air	Day	Feathers, hooked beak, wide wings
Dragonfly	Flying near the ground	Day	Two pairs of wings, jointed legs
Butterfly	Flying near the ground	Day	Coloured wings with scales
Spotted deer	Forest floor	Day	Hair with white spots, hooves
Leopard	Forest floor	Day/unsure	Spotted fur, sharp claws
Tiger	Forest floor	Day/unsure	Striped fur, long canine teeth (long, pointed teeth used for tearing flesh)
Sloth bear	Forest floor	Day/unsure	Long black hair, long claws
Langur / macaque	Tree	Day	Hair, long tail, hands that can grip

Conceptual Learning

Organism	Where do you see it?	When does it appear to be active?	Any visible feature(s)
Rabbit	Forest floor	Day/unsure	Long ears, fur
Frog	Edge of water	Unsure	Wet skin, webbed feet (feet with skin joining the toes)
Crocodile / gharial	Water	Unsure	Scales, long snout
Bat	Flying near trees	Night	Wings made of skin, fur
Python / snake	Tree	Night	Scales, no legs
Porcupine	Forest floor	Night	Quills (sharp spines)
Civet	Forest floor	Night	Spotted fur, striped tail
Slender loris	Tree	Night	Very big eyes, fur
Fireflies	Air near trees	Night	Body that gives light

Table B: Grouping the organisms (Table 12.2 filled)

The grouping criterion	Which organisms fit in this group?	What feature or pattern helped you decide?
Carnivore (animal that eats other animals)	Eagle, tiger, leopard (also owl, python, crocodile)	Eating habits (eat other animals)
Herbivore (animal that eats plants)	Deer, rabbit, langur, porcupine	Eating habits (eat plants)
Can fly	Eagle, owl, bat, peacock, dragonfly, butterfly	Has wings
Active at night	Owl, bat, loris, civet, porcupine, fireflies	Seen on the night side
Live in or near water	Frog, crocodile	Habitat
Body has feathers	Peacock, eagle, owl, small birds	Body covering
Body has hair or fur	Tiger, leopard, deer, bear, langur, rabbit, bat	Body covering
Jointed legs, no backbone	Dragonfly, butterfly, fireflies	Body structure

- **Conclusion:**
  - The same organism can fit into different groups depending on the **criterion** (basis) we choose. Example: the eagle is a carnivore, it can fly, it is active in the day and it has feathers. The bat also flies like the eagle, but it has fur, not feathers.
  - This raises the question of which features scientists should use while grouping living organisms. It leads to the idea of **classification**, a systematic way of organising the Earth's living diversity.
  - Scientists do not group organisms only by habitat or habit, because then unrelated animals come together (the bat and the eagle both fly). So they use basic features that show real relationships.

✓ **Activity 2: Let us read a case study (Pakke Tiger Reserve):**

- **Aim:**
  - To understand, through a case study, why scientists need to classify organisms.
- **Steps:**
  1) Read the case study of Pakke Tiger Reserve carefully.
  2) Think and discuss: how scientists can keep track of so many species, which features can distinguish the four hornbills, and what would happen if the large, old trees disappeared from the forest.

Conceptual Learning

- **The case study:**
  - Pakke Tiger Reserve is a forest in **Arunachal Pradesh**. Scientists have recorded nearly **300 bird species** here.
  - The whole of India has about **1,300 bird species**. So almost one-fourth of India's birds are found in just this one reserve.
  - Pakke supports **four species of hornbills**: Rufous-necked Hornbill, Oriental Pied Hornbill, Great Hornbill and Wreathed Hornbill.
  - These large birds **nest only in large, old trees with suitable cavities** (holes) and **feed on specific fruits**.
  - So, different hornbill species are found in different parts of the forest depending on **tree size** and **fruit availability**.
  - Studying such patterns tells scientists how species are distributed within a forest, which plants and animals are closely linked, and how grouping the four hornbills helps us understand biodiversity.



(a) Rufous-necked hornbill *(b) Oriental Pied hornbill* *(c) Great hornbill* *(d) Wreathed hornbill*

Fig. 24: The four hornbill species of Pakke Tiger Reserve

  - ⇒ **Know these species:**
    - **Rufous-necked hornbill:** Named after its orange-brown (rufous) neck and head, seen clearly in the male. It lives in the hill forests of North East India and is now rare.
    - **Oriental Pied hornbill:** The smallest of the four, with a black and white body. It is the most common one and can live even in disturbed forests.
    - **Great hornbill:** The biggest of the four, with a large yellow and black helmet (casque) on its beak. It is the state bird of Kerala and Arunachal Pradesh.
    - **Wreathed hornbill:** It has ridges like rings (wreaths) at the base of its beak, and the male has a yellow throat pouch (a bag of loose skin under the throat). It flies long distances to find fruiting trees.

- **Observations:**
  - **Keeping track of so many species:** scientists classify them into groups (all hornbills together, then each species separately), give each species a scientific name, and write down its habitat, food and nesting needs in an orderly way.
  - **Telling the four hornbills apart:** by colour of neck and head (orange-brown neck in Rufous-necked; black and white body in Oriental Pied), by size (Great hornbill is the biggest), by the shape and colour of the **casque** (helmet-like part on top of the beak; big yellow casque in the Great hornbill), and by tail pattern, calls, food and nesting trees.
  - **If the large, old trees disappear:** hornbills will have no cavities to nest in and fewer fruits to eat, so their number will decline. Hornbills also spread the seeds of many fruit trees, so those trees will also suffer. The whole forest will be affected.

Conceptual Learning

- **Conclusion:**
  - Classifying the four hornbills is useful as it shows how each species is linked to big fruit trees, since hornbills and these trees depend on each other.
  - So, classification helps scientists keep track of species, tell similar species apart and understand how organisms are linked in a forest.

✓ **Activity 3: Let us study (concept map):**

- **Aim:**
  - To find the criteria that form the basis of the five kingdom classification.
- **Steps:**
  1) Study the concept map (a diagram that shows how ideas are linked) of five kingdom classification.
  2) List the criteria which form the basis of five kingdom classification.
  3) Compare your list with the criteria given.
- **Observations:**
  - The criteria found from the concept map are:

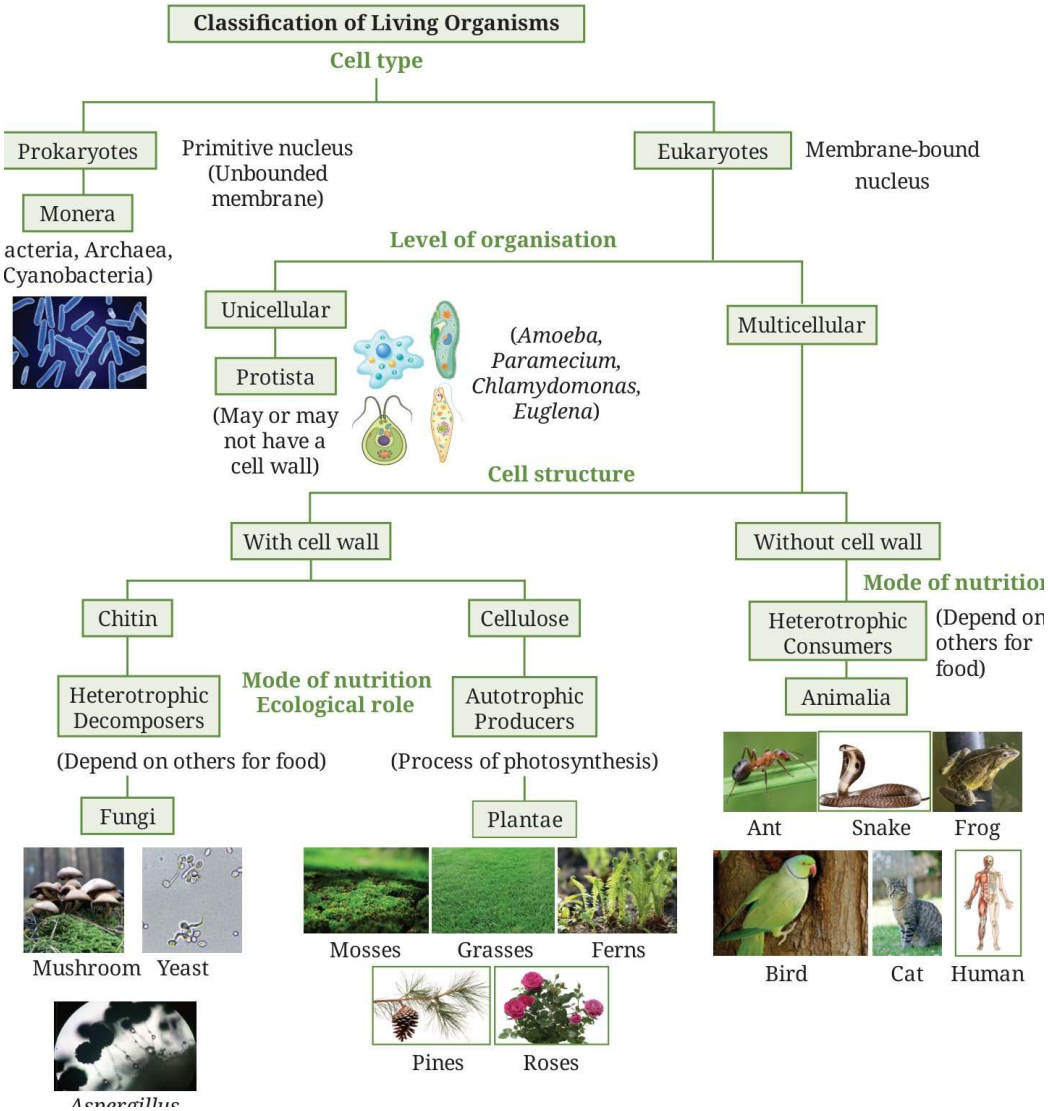


Fig. 25: Concept map of five kingdom classification with examples (Monera: bacteria, archaea, cyanobacteria; Protista: Amoeba, Paramecium, Chlamydomonas, Euglena; Fungi: mushroom, yeast, Aspergillus; Plantae: mosses, grasses, ferns, pines, roses; Animalia: ant, snake, frog, bird, cat, human)

1) **Cell type:** prokaryote (primitive nucleus, not bounded by a membrane) or eukaryote (membrane-bound nucleus). This separates **Monera** from all the others.
2) **Level of organisation:** unicellular or multicellular. This separates **Protista** (unicellular eukaryotes) from Fungi, Plantae and Animalia.
3) **Cell structure:** presence or absence of a cell wall, and what it is made of. Animalia has no cell wall, Fungi has a cell wall of **chitin**, and Plantae has a cell wall of **cellulose**.
4) **Mode of nutrition and ecological role:** autotrophic (producers: Plantae) or heterotrophic (decomposers: Fungi; consumers: Animalia).

- **Conclusion:**
  - Our list matches the given criteria: cell type, cell structure, level of organisation and mode of nutrition.
  - So, the five kingdoms are made on the basis of cell type, level of organisation, cell structure and mode of nutrition.

✓ **Activity 4: Let us explore (bacteria and cyanobacteria):**

- **Aim:**
  - To observe bacteria and cyanobacteria under a microscope.
- **Steps:**
  1) In the school laboratory, observe the available **permanent slides** (ready-made slides) of bacteria and cyanobacteria under the microscope.
  2) Compare them with the figure.

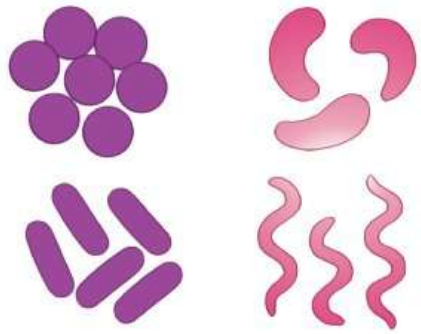


Fig. 26: Members of Kingdom Monera: bacteria of different shapes (round, comma, rod and spiral)

- **Observations:**
  - Very small single cells of different shapes (round, rod, comma and spiral) are seen. No clear nucleus is seen.
- **Conclusion:**
  - Bacteria and cyanobacteria are **single-celled prokaryotes** that are grouped under Monera.

✓ **Activity 5: Let us make (hay infusion):**

- **Aim:**
  - To make a hay infusion (water in which dry grass or straw is soaked) and observe protists in it.
- **Steps:**
  1) Collect some cut grass after the lawn is mowed (cut short), or some straw or fodder (dry food for cattle).
  2) Take a small glass bottle and fill one-fourth of it with the grass, straw or fodder.

3) Fill the bottle with stagnant (still, not flowing) water or pond water and mix it with the plant material.
4) Cover the mouth of the bottle with muslin (thin and soft cotton cloth) cloth and tie it with a thread.
5) Keep the bottle aside undisturbed for a week.
6) Slightly open the mouth of the bottle by removing the muslin cloth, insert a dropper and carefully take a drop of water.
7) Put the drop on a clean slide and observe it under a microscope.

- **Precautions and reasons for the steps:**
  - **Caution:** the hay infusion may smell bad. Wear a lab coat, mask and hand gloves. Discard the hay infusion after **autoclaving** (heating with steam under high pressure) so that all the microbes (microorganisms) are killed.
  - Pond water is used because it already has protists in it.
  - Hay is added as it decays and gives food to bacteria, and many protists feed on these bacteria.
  - The muslin cloth is used because it lets air in but keeps dust and insects out.
  - The bottle is kept for a week as the protists need time to multiply.
- **Observations:**
  - Moving organisms are seen in the drop of water.
  - *Paramecium* is slipper-shaped and moves with hair-like **cilia**.
  - *Amoeba* has no fixed shape and moves with **pseudopodia** (false feet).
  - *Euglena* is green, spindle-shaped and moves with a whip-like **flagellum**.



Fig. 27: Common protists seen in hay infusion: Amoeba, Paramecium, Euglena

- **Conclusion:**
  - Pond water contains many protists like *Amoeba*, *Paramecium* and *Euglena*. They are **unicellular eukaryotes** that live in water or moist places.

✓ **Activity 6: Let us explore (bryophytes):**

- **Aim:**
  - To observe bryophytes and compare their leaf-like parts with normal leaves.
- **Steps:**
  1) Observe some bryophytes with the help of a hand lens (magnifying glass). Collect them in a watch glass (small, round, shallow glass dish).
  2) Put a drop of water on them and observe under a **dissecting microscope**. It shows magnified (enlarged) images of live specimens (samples of organisms used for study).
- **Observations:**
  - **How they differ from normal leaves:** the leaf-like parts of bryophytes are very small and thin, often just one cell thick. They have no veins, no midrib (the main central vein of a leaf) and no stalk. Normal leaves are bigger and thicker, have a network of veins and have a stalk (petiole).

Conceptual Learning

- **Conclusion:**
  - Bryophytes have simple leaf-like structures, not true leaves. They stay small and need moist places because they lack vascular tissues (xylem and phloem) and must absorb water through their whole body surface.

✓ **Activity 7: Let us compare (fern stem and sunflower stem):**

- **Aim:**
  - To compare the vascular tissue in the stem of a fern with that in the stem of a higher plant (sunflower).
- **Steps:**
  1) Recall the cross section of the sunflower stem studied earlier.
  2) Compare the cross section of the stem of a fern (a pteridophytic plant) with the cross section of the sunflower stem.
  3) Write the differences in the vascular tissue of the fern stem and the stem of higher plants. Share and discuss them in class.
- **Observations:**
  - **Fern stem:** xylem and phloem form one central strand. Xylem is in the middle and phloem is around it. There is no cambium (a layer of dividing cells between xylem and phloem that makes the stem thicker), so the stem does not grow thicker.
  - **Sunflower stem (a higher plant):** there are many **vascular bundles** (groups of xylem and phloem lying together) **arranged in a ring**. In each bundle, xylem is on the inner side and phloem on the outer side, with cambium between them.
- **Conclusion:**
  - Both stems have xylem and phloem, but in higher plants the vascular tissue is arranged in a more organised way.

✓ **Activity 8: Let us discuss (monocots and dicots):**

- **Aim:**
  - To group leaves as monocots (plants whose seeds have one seed leaf, called cotyledon) or dicots (plants whose seeds have two seed leaves) on the basis of their shape and venation.
- **Steps:**
  1) Collect different leaves from your surroundings.
  2) Observe their shape and **venation** (pattern of veins).
  3) Group them as monocots or dicots.
  4) Discuss how their structures help them adapt in different conditions to survive.
- **Observations:**

Feature	Monocots	Dicots
Venation	Parallel veins	Net-like veins (reticulate)
Leaf shape	Long and narrow, usually no clear stalk	Broad, usually with a stalk (petiole)
Examples	Grass, wheat, maize, banana, bamboo, sugarcane	Mango, peepal, hibiscus, rose, pea, sunflower
How it helps	Narrow leaves lose less water and bend in strong wind; fibrous roots (many thin roots of similar size growing from the base of the stem) hold the soil	Broad leaves catch more sunlight; tap root (one main thick root that goes deep, with smaller side roots) goes deep to find water

Conceptual Learning

- **Conclusion:**
  - Leaves with parallel venation belong to monocots, and leaves with net-like venation belong to dicots.
  - The shape of leaves and roots helps each group survive in its conditions.

✓ **Activity 9: Let us study (classes of Kingdom Plantae):**

- **Aim:**
  - To study the salient features of each plant group and find their advantages and challenges for survival.
- **Steps:**
  1) Carefully study the **salient features** (main features) of each plant group.
  2) Analyse the salient features, and write the advantages for survival of the group and the exceptions or challenges faced (Table 12.3).
- **Observations:**

Table 12.3 (filled): Classes of Kingdom Plantae with their advantages and challenges

Plant groups and salient features	Advantages of the group for survival	Exceptions / Challenges
Thallophyta • Their body is like a thallus, which allows easy absorption of water and nutrients, and exchange of gases from the surroundings.	• Simple plant body helps survival and its dispersal in water. • Every cell takes in what it needs directly, so no transport system is needed.	• Cannot live on land. • No support tissue, so they dry up fast out of water.
Bryophyta • Unlike algae, began to colonise land but strongly depend on moisture. • Slight differentiation in body parts, but not in root, stem or leaves. • Lack vascular tissues. • Require water for reproduction (male reproductive cells swim).	• Called 'plant amphibians' (live on land but need water to reproduce). • Body adapted to live on moist land. • Rhizoids hold them to rocks and soil, and their mats hold moisture.	• Always need moisture. • Stay small as they lack vascular tissues; cannot reproduce in dry places.
Pteridophyta • True roots, stems and leaves. • Vascular tissues (xylem and phloem) transport water and food. • Still depend on water for reproduction. • Do not produce seeds.	• Live on land. • Transport food and water to all parts of the plant. • Can grow taller than bryophytes.	• Reproduction does not take place without water. • No seeds, so the embryo gets no stored food or protection.
Gymnosperm • Live on land. • Needle-like leaves reduce water loss. • Water is not essential for fertilisation. • Seeds not enclosed in fruits; exposed on cones.	• Leaves adapted for dry conditions. • Do not require water for reproduction. • Form seeds for continuity of life (life going on from one generation to the next). • Can live in cold, dry and snowy places.	• Seeds are not covered in the form of fruits. • Seeds less protected and dispersed mostly by wind; no flowers to attract pollinators.

Conceptual Learning

Plant groups and salient features	Advantages of the group for survival	Exceptions / Challenges
Angiosperm • Well-developed roots, stems and leaves. • Sexual reproduction through flowers. • Seeds enclosed within fruits. • Seeds dispersed through insects, birds, animals, wind or water.	• Produce flowers, fruits and seeds. • Well-developed system for reproduction. • Produce seeds for continuity of life. • Seeds are covered. • Wide seed dispersal, so they occupy almost all environments.	• Reproduction depends on pollination (transfer of pollen from one flower to another) by different agents. • Many complex processes through a well-developed tissue system. • If pollinators (bees, birds) decrease, fewer seeds are made.

- **Conclusion:**
  - Different plant groups present different ways of solving problems, each with advantages and challenges for survival.
  - From algae to angiosperms, plants gradually developed transport tissues, seeds and flowers, which helped them live on land.

Conceptual Learning